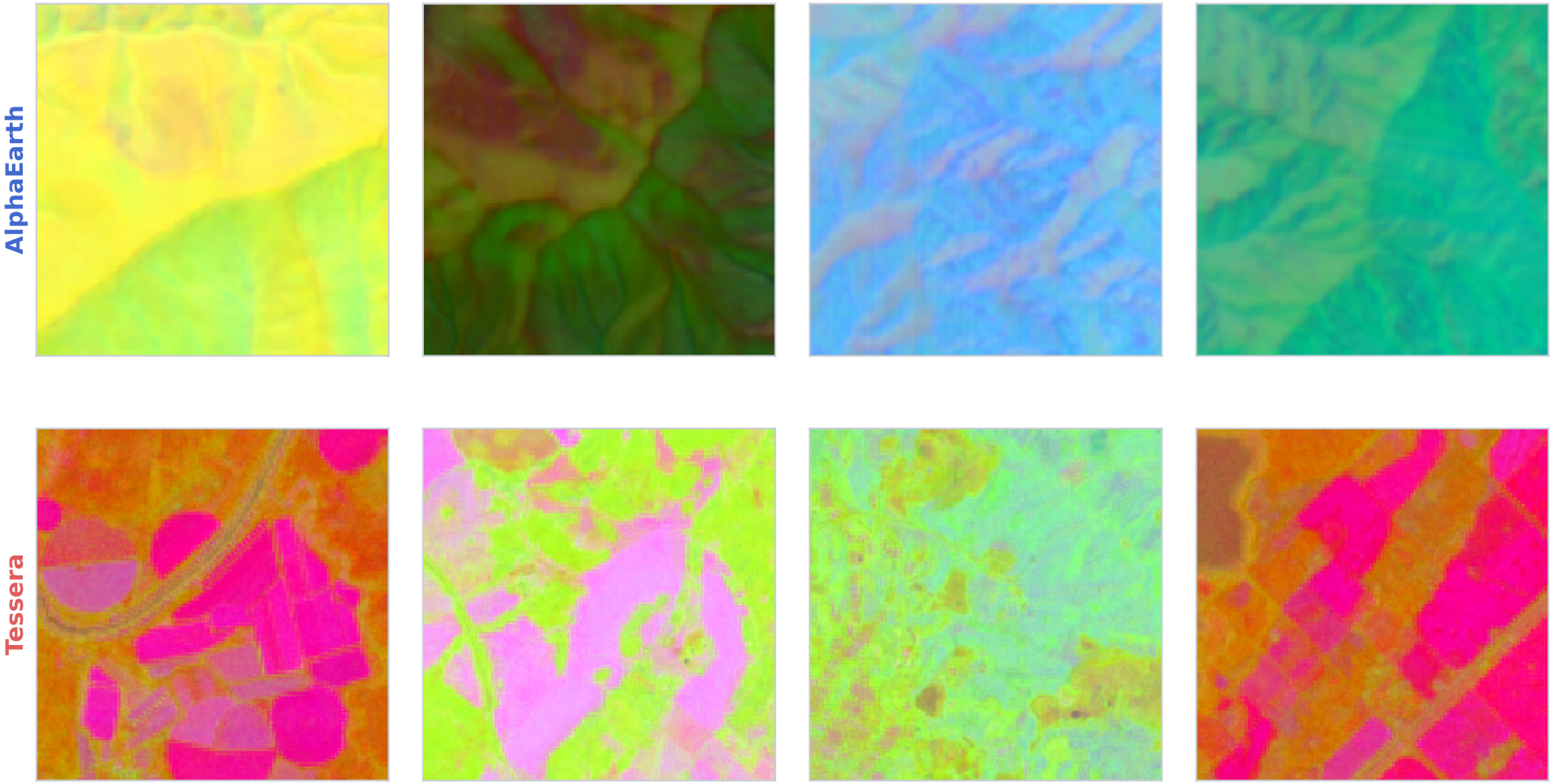
AlphaEarth vs Tessera: two embedding fields, two compression problems

Both datasets are 10 m/px 'embedding fields': every pixel is a vector (AlphaEarth: 64 numbers, Tessera: 128) produced by a foundation model from a year of satellite observations. We compress them with the same codec family. The pages that follow show why the two sources behave very differently under compression.

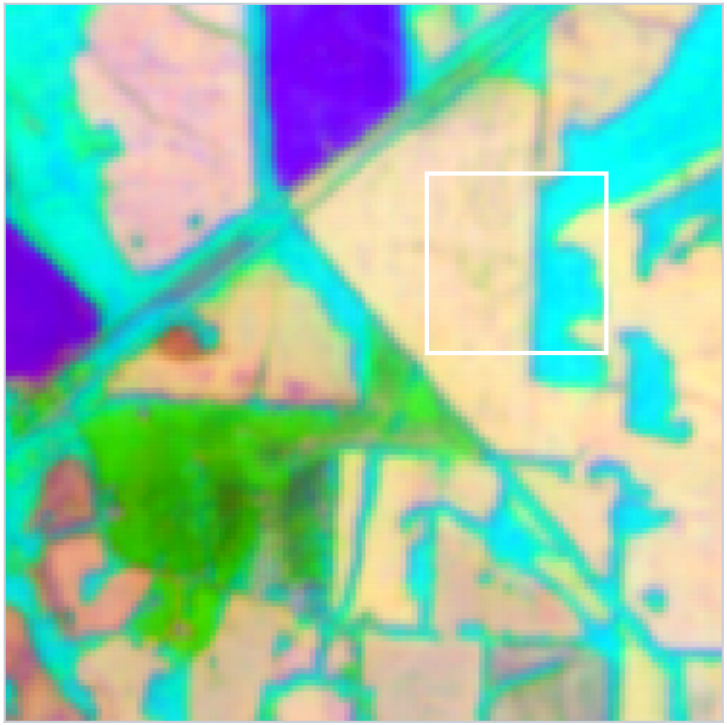


1.9 km windows, embeddings projected to RGB via PCA (colors arbitrary, structure real). Same projection idea for both rows.
Visually: AlphaEarth fields are smooth and region-like; Tessera fields carry much more pixel-scale texture.

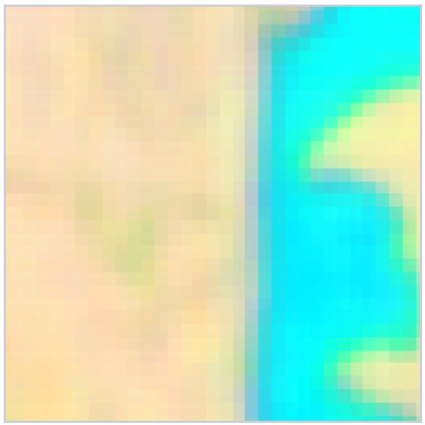
The same 2.6 km of farmland, in both datasets

Chips co-registered by cross-correlating terrain structure (35.7N, 89.1W). Same fields, roads and drainage in both; colors are per-dataset PCA and arbitrary.

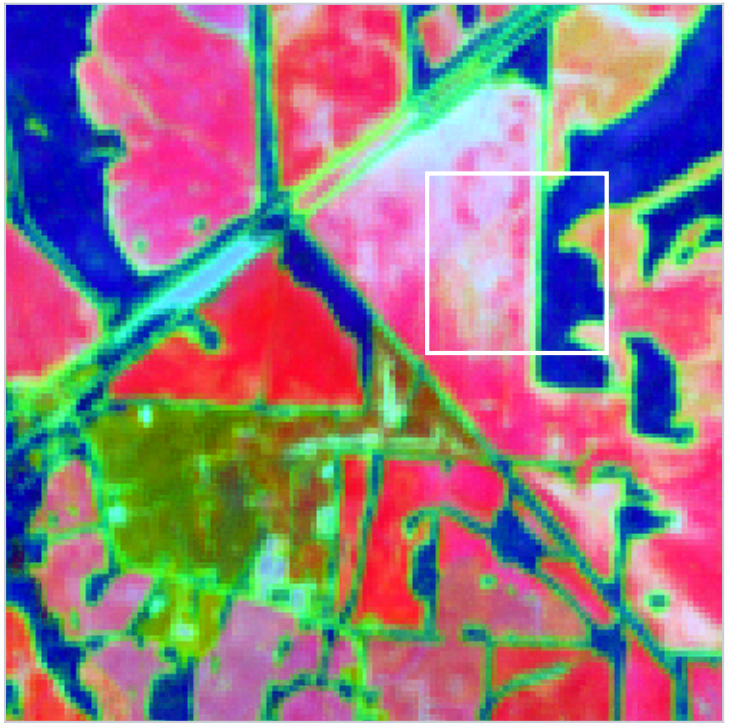
AlphaEarth



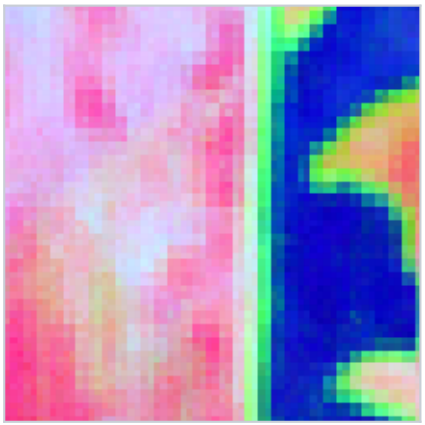
640 m crop



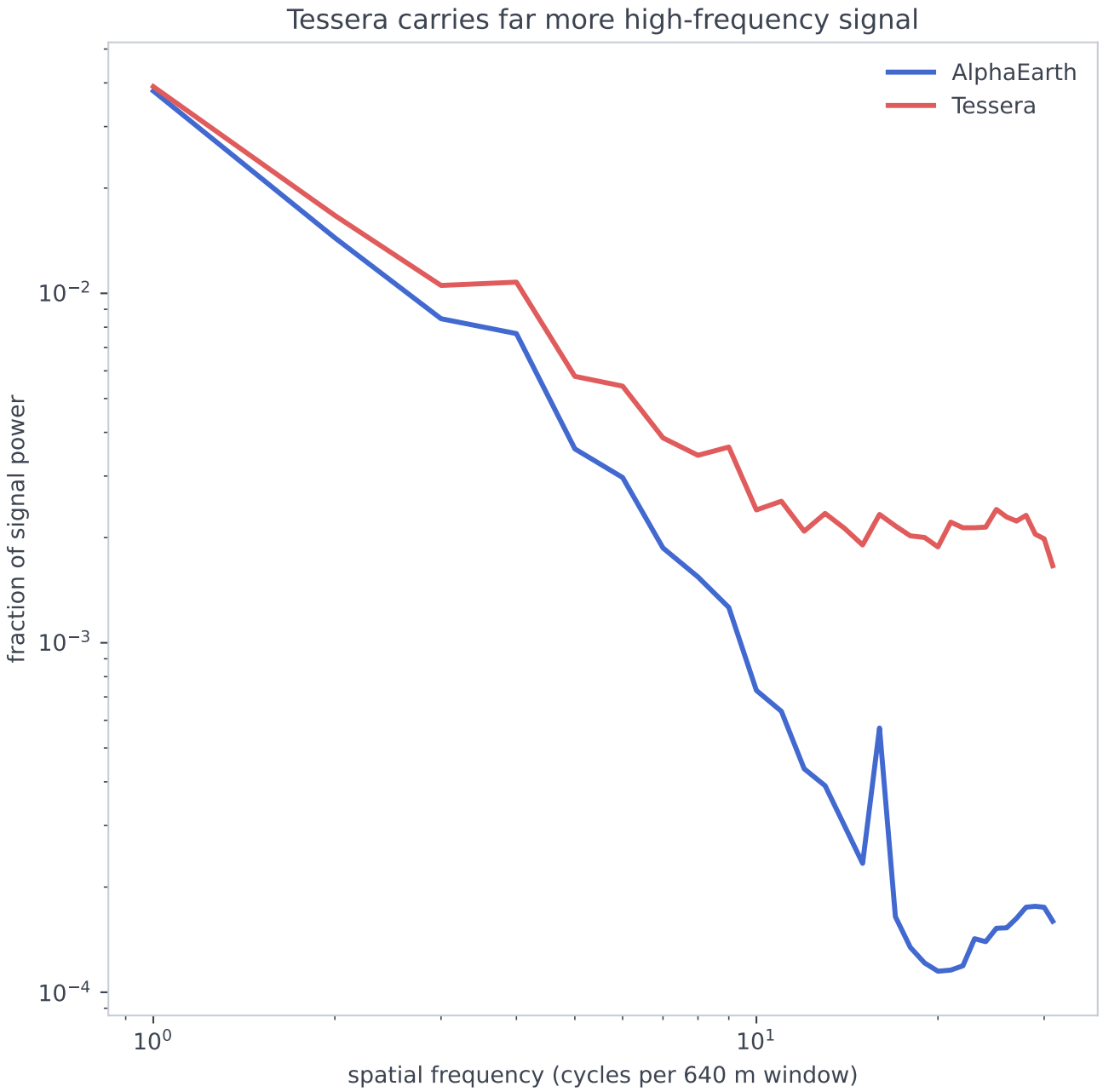
Tessera



640 m crop



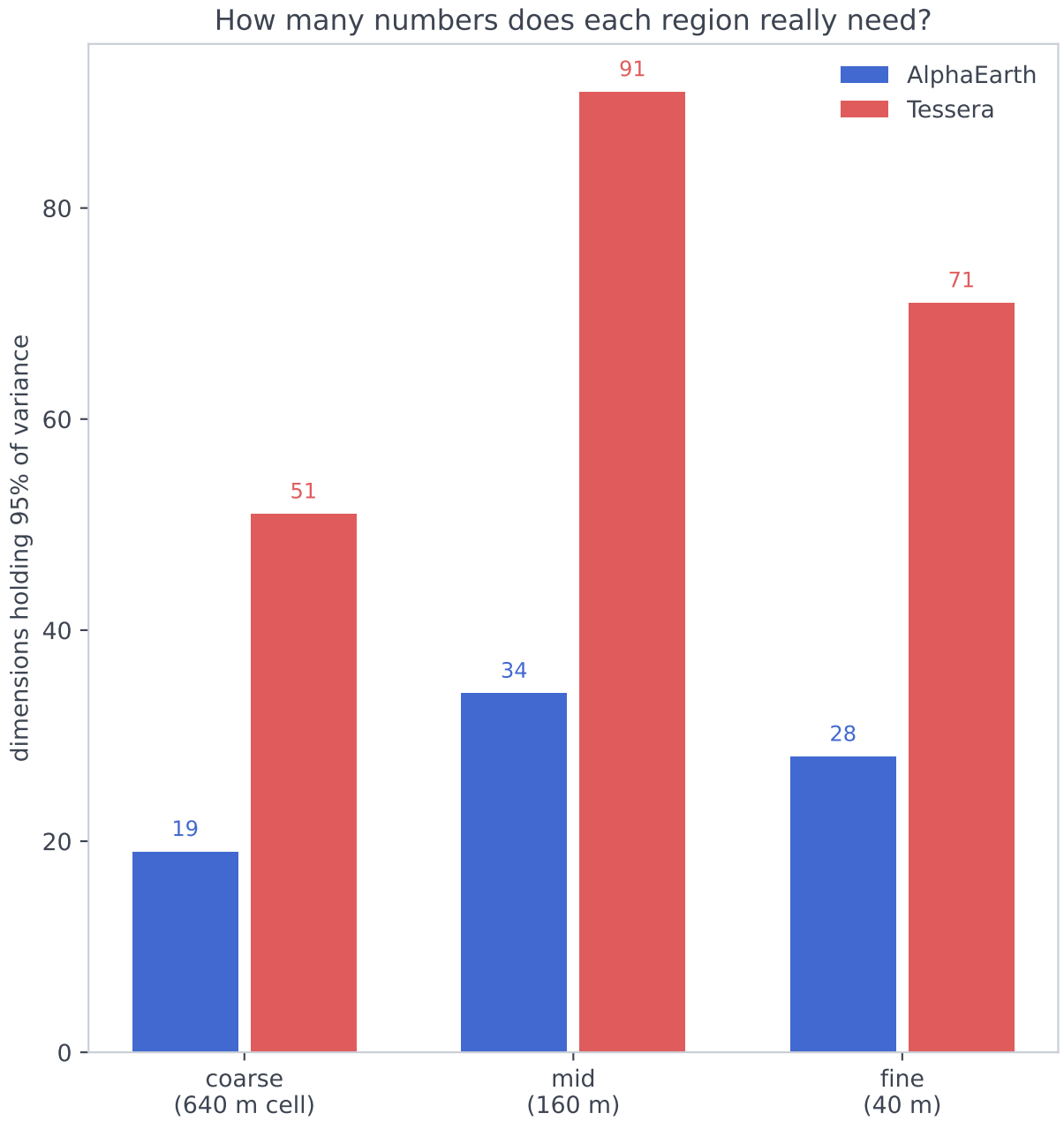
Same location, same zoom: AlphaEarth renders fields as flat regions; Tessera resolves within-field variation down to pixel scale.
That extra texture is real signal the codec must carry — the source of every difference on the following pages.



Power spectra of the embedding fields (share of signal variance by spatial scale).

Fine-scale content (right side) is the expensive part of any compression problem: it cannot be summarized, only stored.

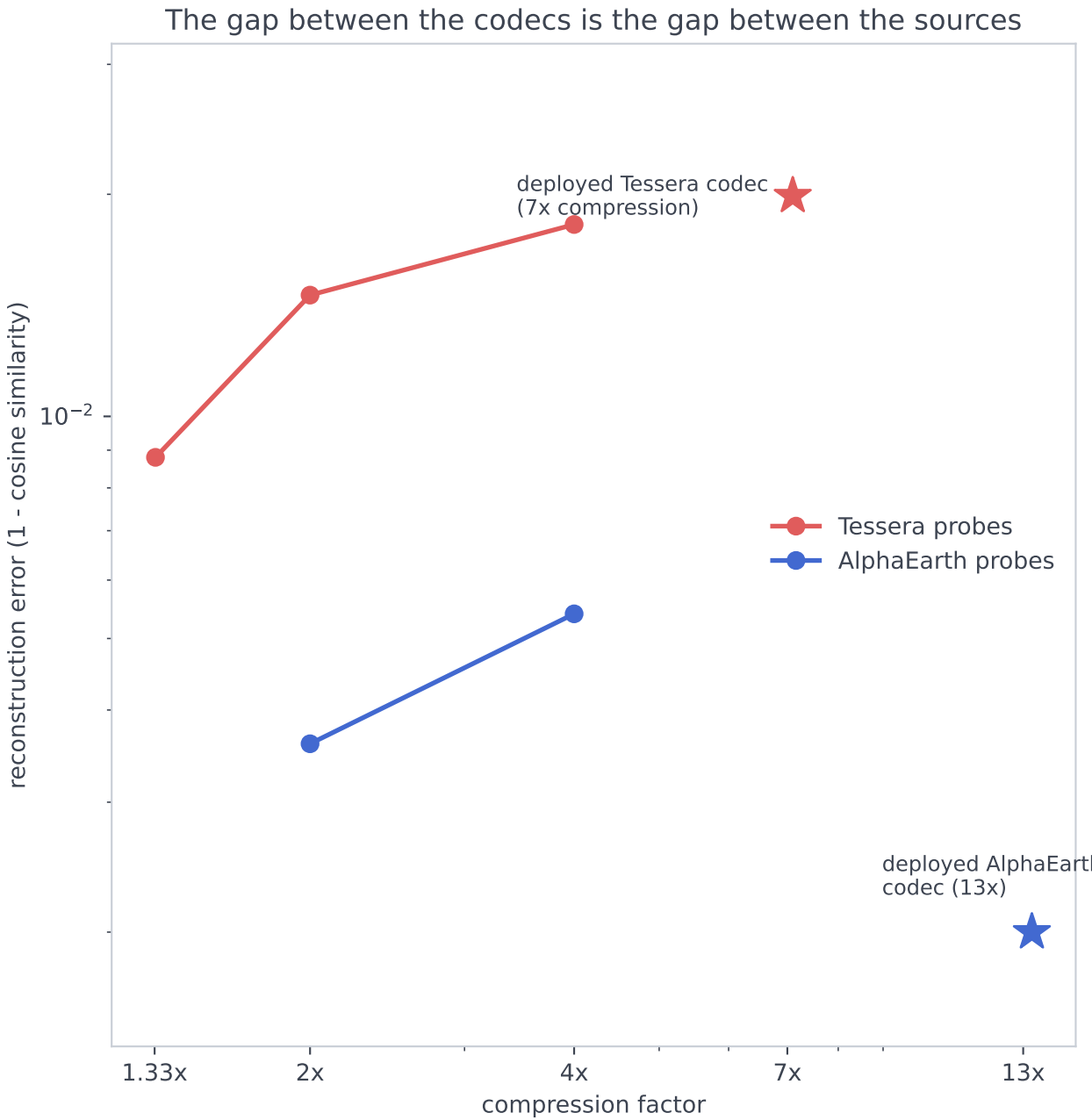
Above ~8 cycles/window Tessera holds roughly an order of magnitude more power than AlphaEarth.



We trained identical multi-scale encoders on both sources and measured the effective dimensionality of the learned summaries.

AlphaEarth content is intrinsically low-dimensional (~20-35 numbers per region). Tessera needs 2-3x more in every tier — its pixels vary in more independent ways.

This is the representation-space view of the spectral difference on the previous page.



Circles: 'best possible' probes — heavyweight models compressing single 4x4-pixel patches with no task other than fidelity.

At every matched rate, Tessera's error is 3.5-4x AlphaEarth's; both fall smoothly with rate (no hard noise wall).

Stars: the actual deployed multi-scale codecs. Both beat their own per-patch bound — spatial context is worth about one compression octave — and each sits where its source's curve says it must.

Takeaways

1. The two sources pose genuinely different compression problems. AlphaEarth fields are smooth and low-dimensional; Tessera fields carry 2-3x more independent variation per pixel and ~10x more fine-scale signal power.

2. The codec is close to optimal on both. Deployed models beat their own context-free per-patch bounds at matched rates — multi-scale context buys roughly one compression octave. The quality difference between the deployed codecs (cos ~0.998 vs ~0.980) is a property of the sources, not of the method.

3. Error is the right axis, not similarity. In error terms (1 - cos, log scale), both sources trade error for rate along smooth, similarly-shaped curves; AlphaEarth's sits ~4x lower. "0.996 vs 0.998" is a factor-of-two difference in error.

4. Roadmap: Tessera quality targets should be set per-rate from its curve (e.g. ~0.99 ceiling requires roughly 2.5-3x naive-rate latents, now in training as a 4-tier model with a 10 m detail tier). The productive engineering axis is rate-at-fixed-quality: quantized codebooks (int8, ~72 MB download), residual VQ ladders, and entropy coding.